Investigation of Delayed Neutron Precursor Data Uncertainties for Microscopic Calculations

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Abstract

This paper presents an analysis on the sensitivity of delayed neutron precursor (DNP) group parameters, half-lives τ_k and yields $\nu_{d,k}$, to uncertainties in their nuclear data using thermal fission of ^{235}U for various datasets: the International Atomic Energy Agency (IAEA) beta-delayed neutron emission database, ENDF/B-VII.1, ENDF/B-VIII.0, JEFF-3.1.1, and JENDL-5. These uncertainty-weighted sensitivities provide a detailed view of which DNPs need more accurate data such that analyses using individual DNPs, such as microscopic group parameter formulation and summation calculations, may be more accurate. Improving the accuracy of these analyses is of particular importance for kinetics models, which rely on accurate DNP group parameters. The microscopic approach for group parameter calculation is useful for providing insight into how each DNP contributes to delayed neutron emission, but it requires accurate nuclear data. The analyses conducted in this work reveal several DNPs that correlate strongly with the DNP group parameters, indicating a need for more accurate data for those DNPs. There are also DNPs which have large uncertainties and vary significantly between datasets which also warrant additional attention. Large discrepancies are shown in ENDF datasets for ^{85}As, ^{86}As, ^{100}Rb, and ^{137}Sb; under prediction of the delayed neutron yield ν_d in JEFF-3.1.1 correlates to many small discrepancies in various DNPs; and JENDL-5 is shown to offer good agreement with experimental and calculated delayed neutron yields.

Keywords — Delayed Neutron Precursors, Group Parameters, Nuclear Data, Nuclear Data Uncertainties, ENDF, JEFF

I. INTRODUCTION

Modeling and simulation of nuclear reactors enables more accurate predictions about the reactor behavior during transient events, such as removal of a control rod. To accurately model these dynamics, several different effects must be included: the change in the neutron density, the various feedback effects that influence the reactivity, the emission of prompt neutrons, and the emission of delayed neutrons. These effects are well understood and have been simplified using various approximations. For example, the delayed neutrons are treated as having six to eight sources, or delayed neutron precursor (DNP) groups, even though in reality there are hundreds of precursor nuclides that decay and emit these delayed neutrons [1].

There are several different methods for calculating these DNP groups, and in this work the microscopic approach is of particular interest [2, 3, 4, 5, 6, 7, 8, 9]. This approach uses experimental data for individual DNPs combined together computationally. One strength of this approach is that it provides insight into how each individual DNP contributes to the DNP group parameters. However, this strength is also a weakness, as it relies on the nuclear data for every DNP individually, which means a few large uncertainties in the various DNPs can lead to uncertain results for the DNP group parameters. The microscopic approach can also be used for calculation of summed parameters such as the total delayed neutron yield ν_d and the average half-life $\bar{\tau}$ directly from the DNP data, generally referred to as conducting a summation, or aggregate, calculation.

There have been investigations in the literature that use the microscopic approach to evaluate sensitivities and compare different datasets [10, 8, 6]. Although these analyses are informative and show many of the differences between datasets, they do not discuss the importance of DNPs in regards to their effect on the calculated DNP group parameters but instead based on their contribution to the total delayed neutron yield. Generally the larger the delayed neutron yield of a DNP, the more important it is, but these analyses do not investigate how the uncertainty in each DNP propagates through to calculated DNP group parameters.

This work seeks to expand upon these previous analyses by using the microscopic approach in combination with Monte Carlo stochastic sampling of nuclear data uncertainties to evaluate uncertainty-weighted sensitivities of individual DNPs to DNP group half-lives and delayed neutron yields. These results indicate which DNPs in different datasets have large impacts on the DNP group parameters, as well as which DNPs have both large impacts and large uncertainties relative to those impacts. By providing the data of importance for these datasets, future analyses can target experiments to produce those specific data to further improve DNP group parameters and kinetics models.

II. METHODS

This section details the three stages to the analyses presented in this paper: the data treated as an input, the solver which calculates the data of interest, and the data analysis which details how the solver results are processed. The solver itself can be considered a converter of the DNP nuclear data into six group DNP parameters, which are then passed into data analysis calculations.

II.A. DNP Nuclear Data Treatment

There are two different sets of data used in this work: the nuclear data for each individual DNP and the calculated DNP group parameters that approximate those individual DNPs. The I DNPs have the following data:

$$
\begin{aligned}
I &= \text{the total number of DNP [-]} \\
CFY_i &= \text{the cumulative fission yield of DNP } i \text{ [-]} \\
P_{n,i} &= \text{the emission probability of DNP } i \text{ [-]} \\
\tau_i &= \text{the half-life of DNP } i \ [s] \\
\chi_{d,i} &= \text{the delayed neutron energy spectrum from DNP } i \text{ [-]}.
\end{aligned}
$$

The cumulative fission yield represents the total production of that nuclide for a single fission, including decays that eventually lead to that nuclide. The emission probability is the probability for a delayed neutron to be emitted from a given DNP[a]. For this work, the delayed neutron energy spectra are not investigated, as it is considered out of scope.

[a]This can alternatively be considered the number of delayed neutrons emitted per radioactive decay.

The DNP group parameters have the following data:

$$\nu_{d,k} = \text{the delayed neutron yield of group } k \text{ [-]}$$
$$\tau_k = \text{the half-life of group } k\ [s]$$
$$\chi_{d,k} = \text{the delayed neutron energy spectrum from group } k \text{ [-].}$$

The delayed neutron yield for each group $\nu_{d,k}$ combines the emission probabilities and cumulative fission yields, representing the number of delayed neutrons emitted from group k from a single fission.

The datasets used in this work are given in Table I, including two different Evaluated Nuclear Data File (ENDF) datasets, the beta-delayed neutron emission database from the International Atomic Energy Agency (IAEA), and Japanese Evaluated Nuclear Data Library (JENDL)-5. The IAEA beta-delayed neutron emission database does not have cumulative fission yield data combined with it, though analyses using the data use Joint Evaluated Fission and Fusion (JEFF)-3.1.1 cumulative fission yields [11]. For the analyses in this work, the cumulative fissions yields from all of the datasets will be used and compared against to offer a comprehensive analysis.

TABLE I
Various data sources as well as the data which they contain.

			Available Data		
Data name	Year	Source	τ_i	$P_{n,i}$	CFY_i
ENDF/B-VII.1	2011	[12]	Yes	Yes	Yes
JEFF-3.1.1	2017	[13]	Yes	Yes	Yes
IAEA database	2018	[11]	Yes	Yes	No
ENDF/B-VIII.0	2018	[14]	Yes	Yes	Yes
JENDL-5	2021	[15]	Yes	Yes	Yes

The data of interest in these datasets are for the individual DNP nuclear data. This data is passed into the solver, which then converts these data into DNP group parameters for analysis and post-processing.

II.B. Microscopic and Summation Calculations

Within the microscopic approach, there are two distinct processes that can be used. The summation calculation is the simpler of the two, and simply involves summing the nuclear data to calculate the total delayed neutron yield and average half-life. The full microscopic approach is used to calculate the DNP group parameters, which give time-dependency and can be used to calculate the total delayed neutron yield and average half-life from the DNP group parameters as well as the DNP nuclear data.

Equations (1) and (2) are the equations for the total delayed neutron yield and average half-life, respectively, from the summation calculation as:

$$\nu_d = \sum_{i=1}^{I} P_{n,i} CFY_i, \tag{1}$$

and

$$\bar{\tau} = \frac{1}{\nu_d} \sum_{i=1}^{I} CFY_i P_{n,i} \tau_i. \tag{2}$$

The uncertainties for each calculation are propagated as shown in Equations (3) and (4) as:

$$\Delta \nu_d^2 = \sum_{i=1}^{I} (\Delta P_{n,i} CFY_i)^2 + (P_{n,i} \Delta CFY_i)^2, \tag{3}$$

and

$$\begin{aligned}\Delta \bar{\tau}^2 = \frac{1}{\nu_d^2} \sum_{i=1}^{I} &(\Delta CFY_i P_{n,i} \tau_i)^2 + (CFY_i \Delta P_{n,i} \tau_i)^2 \\ &+ (CFY_i P_{n,i} \Delta \tau_i)^2 + \frac{1}{\nu_d^2} (CFY_i P_{n,i} \tau_i \Delta \nu_d)^2.\end{aligned} \tag{4}$$

These calculations are automated in the solver using the uncertainties Python package [16]. The standard deviation propagated in the summation calculation are given by the dataset used, and any values without a standard deviation or a standard deviation of 0 use a value of 10^{-12}.

The microscopic approach enables calculation of time-dependent parameters by calculating the delayed neutron count rate from each DNP after irradiation of a fissile nuclide. The solver first calculates the concentration of each DNP, followed by the delayed neutron count rate, and finally uses the SciPy optimize module to calculate the six DNP group parameters that optimally fit that delayed neutron count rate [17]. To calculate sensitivities and propagate uncertainties, a Monte Carlo sampling technique is used [8]. The solution is calculated with tolerances of 10^{-11} for ten randomly sampled initial conditions for the nominal nuclear data, which is then passed as an initial guess for each of the sampled Monte Carlo solves. The concentrations have the uncertainty propagated using the uncertainties package, and then in the count rate calculation the concentration N_i, half-life τ_i, and emission probability $P_{n,i}$ are sampled using a normal distribution with a different random value for each DNP.

The concentration of each DNP is calculated assuming the sample has reached equilibrium[b] [7]:

$$N_i(t = \infty) = \frac{CFY_i}{\lambda_i}, \tag{5}$$

where

$$\lambda_i = \text{the decay constant for DNP } i \; [s^{-1}].$$

The delayed neutron count rate can then be calculated as shown in Equation (6) as:

$$\dot{n}_d(t) = \sum_{i=1}^{I} P_{n,i} N_i(t = \infty) e^{-\lambda_i t}. \tag{6}$$

The count rate is first calculated using the nominal nuclear data before this calculation is repeated using the independently, normally sampled nuclear data for each DNP.

Once the count rate is calculated, a non-linear least squares calculation can be used to determine the combination of DNP group parameters most optimally fits the calculated delayed neutron count rate. The coefficient matrix and solution vector for the non-linear least squares solve are based on the DNP group parameter fit for the delayed neutron count rate, given in Equation (7) as [4]:

$$\dot{n}_d(t) = \sum_{k=1}^{K} \nu_{d,k} e^{-\lambda_i t}. \tag{7}$$

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays [1, 18, 19]. Equation (8) gives the

[b] The parasitic absorption and transmutation cross-sections of the DNPs are negligible and are not included in this equation.

general form of the least squares equation as:

$$A\vec{x} \approx \vec{b}, \tag{8}$$

where

$$A = \text{the coefficient matrix [-]}$$
$$x = \text{the solution vector } [\# \cdot s^{-1}]$$
$$b = \text{the target vector } [\# \cdot s^{-1}].$$

Equation (9) gives the coefficient matrix as:

$$A = \begin{bmatrix} e^{-\lambda_1 t_0} & e^{-\lambda_2 t_0} & \dots & e^{-\lambda_K t_0} \\ e^{-\lambda_1 t_1} & e^{-\lambda_2 t_1} & \dots & e^{-\lambda_K t_1} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-\lambda_1 t_m} & e^{-\lambda_2 t_m} & \dots & e^{-\lambda_K t_m} \end{bmatrix}. \tag{9}$$

The solution vector contains only the group yields, given in Equation (10) as:

$$\vec{x} = \begin{bmatrix} \bar{\nu}_{d,1} \\ \bar{\nu}_{d,2} \\ \vdots \\ \bar{\nu}_{d,K} \end{bmatrix}. \tag{10}$$

The target vector contains the delayed neutron count rate data for m time-steps, shown in Equation (11) as:

$$\vec{b} = \begin{bmatrix} \dot{n}_d(t_0) \\ \dot{n}_d(t_1) \\ \vdots \\ \dot{n}_d(t_m) \end{bmatrix}. \tag{11}$$

Using least squares with a non-relative residual would lead to early time-step count rates dominating due to their larger magnitudes. This is resolved by minimizing a relative residual, given by Equation (12) as:

$$r = \min_x \left|\left|\frac{\vec{b} - A\vec{x}}{\vec{b}}\right|\right|_2, \tag{12}$$

where the Euclidean norm, $||\cdot||_2$, is defined in Equation 13 as:

$$||\vec{x}||_2 = \sqrt{x_1^2 + \cdots + x_n^2}. \tag{13}$$

The relative residual is used instead of a chi-squared statistic because the Monte Carlo sampling method is already in place to calculate and propagate the uncertainties [20].

The DNP group parameters can then be used to evaluate the total delayed neutron yield and average half-life, similar to Equations (1) and (2). This is shown in Equations (14) and (15) as:

$$\nu_d \approx \sum_{k=1}^{K} \nu_{d,k}, \tag{14}$$

and

$$\bar{\tau} \approx \frac{1}{\nu_d} \sum_{k=1}^{K} \nu_{d,k}\tau_k. \tag{15}$$

These equations are approximate because the number of groups K used is fewer than the total number of DNPs I. To differentiate between the calculation of these parameters using I DNPs and K groups, the parameters will be written as $\nu_d(I)$ and $\bar{\tau}(I)$ when calculated using I DNPs and $\nu_d(K)$ and $\bar{\tau}(K)$ when calculated using K groups.

II.C. Sensitivity and Uncertainty Analysis

The Pearson correlation coefficient (PCC) is used to analyze the DNP group parameters calculated from the solver for various unique combinations of sampled DNP nuclear data and is calculated using the SciPy stats module [17]. The PCC measures the linear correlation between the DNP data and the calculated DNP group parameters. A measure of 0 indicates there is no correlation, a measure of +1 indicates perfect positive linear correlation, and a measure of −1 indicates perfect negative linear correlation. For example, ^{87}Br can be expected to have a PCC near +1 for the impact of its half-life τ_i on the half-life of group one[c] τ_1. This is because ^{87}Br is

[c] Group one is the longest-lived DNP group.

the longest-lived DNP[d], so the longest-lived DNP group will be highly dependent on its nuclear data.

The PCC was selected to measure variation in the group parameters because the sensitivities did not generally have strong non-linear behavior. The non-linear least squares fit will result in non-linear effects, but linear correlations dominate in the observed sensitivities. The PCCs are calculated for each DNP group half-life τ_k and delayed neutron yield $\nu_{d,k}$ based on changes in each DNPs' concentration N_i, half-life τ_i, and emission probability $P_{n,i}$. The PCC for each specific combination can be represented as $PCC_{i,v,k}$, where i represents the DNP i; v represents the DNP value, such as the half-life, emission probability, or concentration; and k represents the DNP group.

To improve the analysis in this work, the $PCC_{i,v,k}$ data was combined to form more concise coefficients. The total PCC for a specific DNP i is given in Equation (16) is represented as:

$$PCC_i = \sum_v \sum_k |PCC_{i,v,k}|, \tag{16}$$

where

v = the DNP value, representing the half-life, emission probability, or concentration of DNP i [-]

$PCC_{i,v,g}$ = the PCC for DNP i and value v with DNP group parameter k [-].

These PCCs can be combined with the relative uncertainty associated with each DNP to capture the nuclides which both impact the DNP group parameters most significantly and have large uncertainties. Specifically, this is calculated in Equation (17) as the relative uncertainties times the magnitude of the PCC for a DNP i and each DNP value v as U_i:

$$U_i = \sum_v \frac{\sigma_{i,v}}{x_{i,v}} \sum_k |PCC_{i,v,k}|, \tag{17}$$

[d]There are other DNPs with longer half-lives, such as ^{211}Tl, ^{210}Tl, and ^{210}Hg, but they live only tens of seconds longer and generally have negligible concentrations.

where

$U_i =$ the total uncertainty-weighted sensitivity from the contribution of every DNP i [-]

$\sigma_{i,v} =$ the uncertainty in the DNP value v for DNP i [-]

$x_{i,v} =$ the DNP value v, representing the half-life, emission probability, or concentration of DNP i [-].

Equation (17) shows that the relative uncertainty for each nuclear data component that is randomly sampled, such as the emission probability, is multiplied by the summed magnitudes of the PCC for each DNP group parameter using that data. These values are then all summed, giving a total measure of the relative uncertainty-weighted PCC for each DNP, which indicates how much attention that specific DNP needs. This metric is larger when:

- changes in the DNP's nuclear data cause large changes in the DNP group parameter(s)
- the relative uncertainty in the DNP's nuclear data is large.

This can also be represented for any individual DNP value by removing the summation over v. This is shown in Equation (18) which gives the uncertainty-weighted PCC for a given DNP i and nuclear data v as:

$$U_{i,v} = \frac{\sigma_{i,v}}{x_{i,v}} \sum_k |PCC_{i,v,k}| . \tag{18}$$

III. RESULTS

This section presents the results for different datasets of cumulative fission yields, emission probabilities, and half-lives. Each sub-section will contain the following analyses for each dataset:

- the total delayed neutron yield calculated from summation of DNPs $\nu_d(I)$ and from the group parameters $\nu_d(K)$
- the average half-life calculated from summation of DNPs $\bar{\tau}(I)$ and from the group parameters $\bar{\tau}(K)$
- the six DNPs with the largest yield contributions

- the tabulated ten largest PCCs and uncertainty-weighted PCCs
- a chart of nuclides made using The Advanced Reactor Modeling Interface (ARMI) with summed PCCs and uncertainty-weighted PCCs [21]
- a bar chart with the most significant uncertainty-weighted PCCs.

Table II shows the default parameters used for each simulation.

TABLE II
Default parameters for each microscopic calculation.

Parameter	Value	Units
Fissile Nuclide	^{235}U	[-]
Incident Neutron Energy	0.0253	$[eV]$
Final Decay Time	600	$[s]$
Decay Time Spacing	Logarithmic	[-]
Decay Time Nodes	800	[-]
Sampled Simulations	5000	[-]

As shown in Table I, the IAEA database does not include cumulative fission yield data. For that analysis, the JENDL-5 cumulative fission yields are used since they are the most recent data. However, part of the analysis will also investigate the effects of hybrid datasets that vary the emission probability, half-life, and cumulative fission yield data.

This work is also verified with total delayed neutron yields from a related IAEA Coordinated Research Project (CRP) analysis [11]. The analysis presented a benchmark problem using JEFF-3.1.1 cumulative fission yields with ENDF/B-VIII.0 emission probabilities and half-lives, which is replicated in this work [22, 14]. The four groups in the study are The French Alternative Energies and Atomic Energy Commission (CEA), The Centre for Energy, Environmental and Technological Research (CIEMAT), the Japan Atomic Energy Agency (JAEA), and the University of Nantes, all of whom reported the same result of 0.01609 ± 0.0008 for the total delayed neutron yield from thermal fission of ^{235}U. This agrees with the same result calculated in this work of 0.01609 ± 0.0008. This work is also verified with the total delayed neutron yield ν_d from Leconte et al. from JEFF-3.1.1 cumulative fission yields combined with ENDF/B-VII.1 emission probabilities and half-lives. Specifically, this work calculates a value of 0.0157 ± 0.0008, which exactly agrees with the results from Leconte et al. [23]. Finally, there is also agreement with Huynh et al. using the JEFF-3.1.1 data to calculate the total delayed neutron yield of 0.0148 ± 0.0008 [7]. The verification uses the results from summation calculations, but each section demonstrates that the summation calculation results offer agreement within uncertainty bounds between the DNP group parameter evaluated values and the summation calculation values.

For the first set of results, additional information will be included that is not included in the other sections. Specifically, the delayed neutron count rate will be provided as well as a comparison to the count rate from calculated from other DNP group parameters in the literature. There will also be an example figure demonstrating how the stochastic sampling affects the DNP group parameters. Additionally, specific DNP examples will be provided demonstrating the PCCs.

III.A. IAEA Beta-Delayed Neutron Emission Database

Figure 1 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). This equation shows that the largest yield contributors have large emission probabilities P_n and cumulative fission yields CFY. This doesn't necessarily indicate there will be a large concentration of the DNP, as it may be short lived, but rather for any given fission, that DNP has a higher probability to form and emit a delayed neutron. The total delayed neutron yield for this dataset is 0.0160 delayed neutrons per fission event. There are a total of 220 DNPs in this dataset, and five of them capture over 50% of the total delayed neutron yield.

Figure 2 shows the histogram for the sampled group one half-lives τ_1 as well as how the delayed neutron count rate compares to those using DNP group parameters from the literature [24, 5, 25]. The histogram shows that the group parameters follow a normal distribution as the nuclear data for each DNP is randomly sampled, with the group half-life as the mean of that normal distribution. Although not shown, the other group parameters follow this same trend. The Keepin normalized delayed neutron count rate shows how the delayed neutron count rate calculated in this work, both for every count rate from the sampled data and from the group fit using the calculated DNP group parameters compare. A value of 1 at any given time indicates the delayed neutron count rate at that time perfectly matches the delayed neutron count rate from the Keepin et al. DNP group parameters. These results show that the Keepin et al. DNP group parameters agree

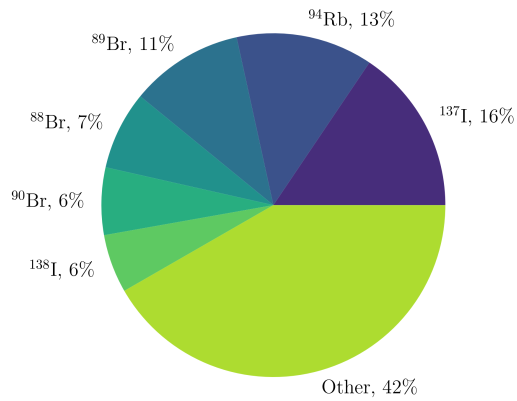
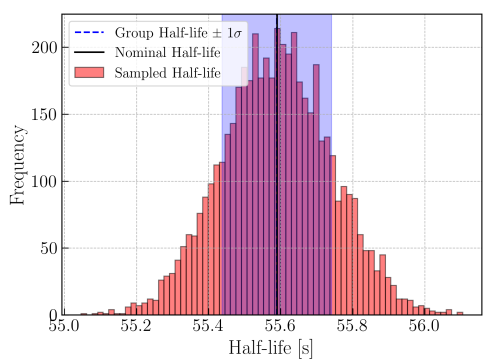
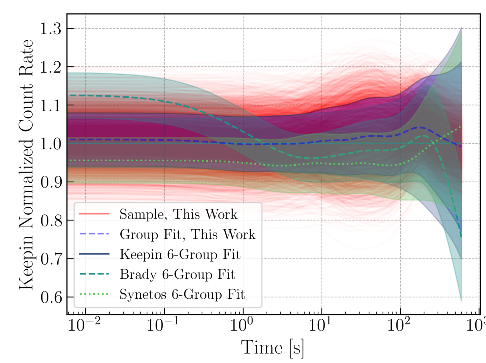


Fig. 1. The fraction of the delayed neutron yield from particular DNPs from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities. The five DNPs with the largest delayed neutron yield contributions supply over 50% of the total delayed neutrons per fission event. The remaining 215 DNPs provide approximately 48% of the total delayed neutron yield.

well with group parameters calculated using the IAEA beta-delayed neutron emission database and JENDL-5 cumulative fission yields, with differences smaller than those from Brady and England and from Synetos and Williams.



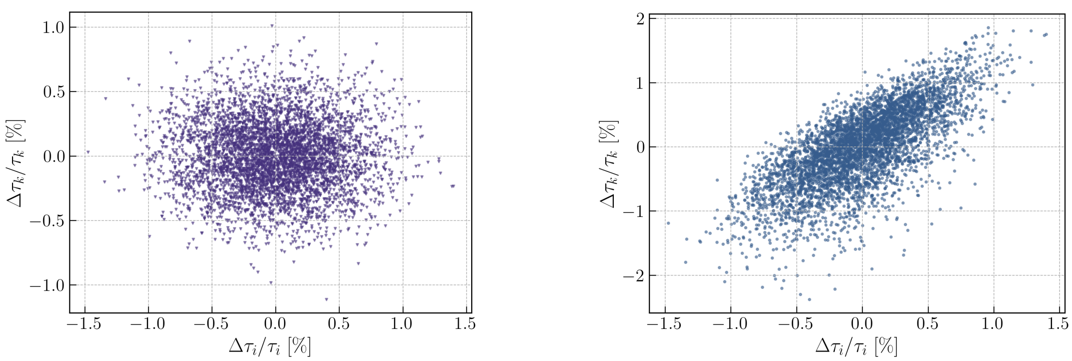
(a) The DNP group one half-life τ_1 evaluated based on the mean of the sampled data.



(b) The delayed neutron count rate as a function of time relative to the Keepin et al. DNP group parameter delayed neutron count rate for DNP six group parameters from this work, Brady and England, and Synetos and Williams [24, 5, 25].

Fig. 2. These plots show how the 5000 samples lead to a distribution of DNP group parameters and delayed neutron count rates from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities. The DNP group one half-life τ_1 shows a normal distribution about a mean, while the Keepin et al. normalized count rate shows this work offers good agreement with the literature.

Figure 3 shows an example for how the PCCs are calculated using the 5000 samples. Each sample is represented as a single point at which the nuclear data is sampled from a normal distribution. Specifically for this figure, the samples are plotted as a function of the ^{137}I half-life. Figure 3(a) shows how the group one DNP half-life τ_1 changes as a function of the ^{137}I half-life. There is no clear increase or decrease in the group one half-life as the ^{137}I half-life varies, which is why the PCC is 0.005 for that combination. However, Figure 3(b) shows how the group two DNP half-life τ_2 has a much stronger correlation, resulting in a PCC of 0.73. These PCC values are calculated for the emission probability, concentration, and half-life for each group for each DNP.



(a) The relative change in the group one half-life τ_1 due to relative change in the ^{137}I half-life with a PCC of 0.005.

(b) The relative change in the group two half-life τ_2 due to relative change in the ^{137}I half-life with a PCC of 0.726.

Fig. 3. These plots show the sensitivity of the DNP group one and two half-lives to the ^{137}I half-life. The group one half-life τ_1 does not correlate with the ^{137}I half-life, resulting in a sample distribution with no discernible slope. The group two half-life τ_2 has a strong, positive linear correlation with the ^{137}I half-life with a PCC of 0.73. This indicates that changes to the ^{137}I half-life will have greater effects on the group two half-life than the group one half-life.

Table III shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general behavior of the delayed neutrons.

Table IV shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k. This table shows that bromine alone has five isotopes that strongly affect the DNP group parameters, with ^{87}Br being the dominant isotope due to its long-half life that makes it the primary DNP for the group one parameters. Generally, DNPs will have a large PCC_i for one of two reasons: they affect several parameters moderately,

TABLE III
IAEA beta-delayed neutron emission database and JENDL-5 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0160 ± 0.0009	$[-]$
$\nu_d(K)$	0.0160 ± 0.0008	$[-]$
$\Delta\nu_d$	0.03 ± 122.82	$[pcm]$
$\bar{\tau}(I)$	9.1 ± 0.5	$[s]$
$\bar{\tau}(K)$	9.0 ± 0.4	$[s]$
$\Delta\bar{\tau}$	0.0 ± 0.6	$[s]$

TABLE IV
IAEA beta-delayed neutron emission database and JENDL-5 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{87}Br	3.678703
^{88}Br	3.397623
^{137}I	3.325644
^{96}Rb	3.277932
^{97}Rb	2.547497
^{94}Rb	2.157988
^{138}I	2.126428
^{89}Br	2.111075
^{90}Br	1.768819
^{91}Br	1.670366

such as ^{96}Rb which has an impact on group six as well as a smaller impact on groups five, four, three, and two as well; or they have a half-life that causes them to have a large impact on one specific group, such as ^{87}Br, which primarily affects group one.

Table V shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k. These ten DNPs do not overlap at all with Table IV, indicating that the JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database have small relative uncertainties for the DNPs in Table IV as otherwise those DNPs would dominate Table V. Interestingly, it appears that cobalt has several isotopes that have large U_i values, indicating that cobalt has large relative uncertainties in its nuclear data relative to its impact on the DNP group parameters.

Figure 4 gives a more detailed view on which DNP values specifically lead to the large U_i values. This offers slightly more insight than Table V, which sums over the various DNP values. This figure shows that the half-lives τ and emission probabilities P_n are the primary drivers of U_i, at least for the ten most impactful DNPs. This indicates that the JENDL-5 cumulative fission

TABLE V
IAEA beta-delayed neutron emission database and JENDL-5 nuclear data ten largest uncertainty-weighted PCCs.

Nuclide	U_i
^{105}Zr	0.281379
^{72}Co	0.268531
^{116}Ru	0.260299
^{105}Y	0.246537
^{75}Co	0.244795
^{68}Co	0.235227
^{116}Rh	0.231495
^{71}Co	0.229786
^{99}Kr	0.226118
^{70}Co	0.222885

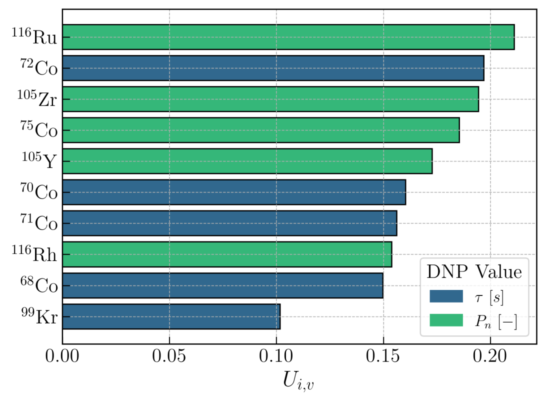
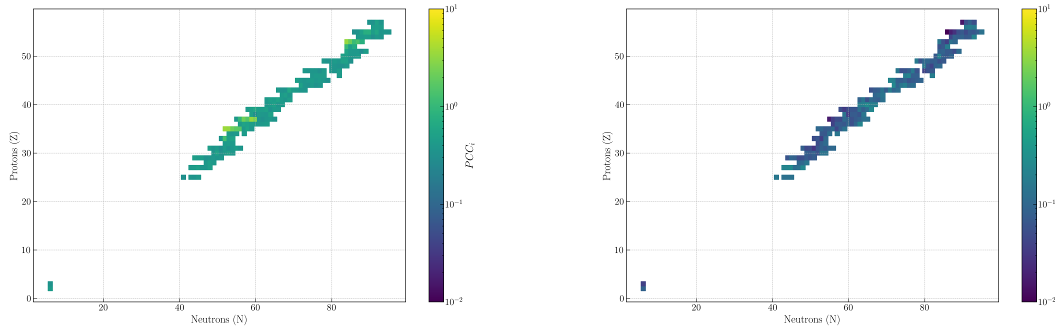


Fig. 4. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities.

yields offer small relative uncertainties compared to the relative uncertainties from the IAEA beta-delayed neutron emission database for DNPs that have more significant effects on the DNP group parameters.

Figure 5 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i, because all of the relative uncertainties are less than 1. There do not appear to be any clear trends, though in Figure 5(a), some horizontal bands of larger PCC_i values can be seen for 35 protons, 37 protons, and 53 protons; these correspond to bromine, rubidium, and iodine, respectively.



(a) The PCC_i, or total PCC, calculated for each DNP.

(b) The U_i, or total uncertainty-weighted PCC, for each DNP.

Fig. 5. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database emission probabilities and half-lives.

III.B. ENDF/B-VII.1

Figure 6 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). These yields can be compared to those in Figure 1, which shows that in ENDF/B-VII.1, the yields of the top DNPs are relatively smaller. There is also the presence of ^{85}As, which did not appear in the top six DNPs when using the IAEA beta-delayed neutron emission database and JENDL-5 nuclear data.

Table VI shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general behavior of the delayed neutrons. However, the average half-life differs by approximately two seconds from Section III.A, while the total delayed neutron yield differs by approximately 300 pcm.

Table VII shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k. These results show that several DNPs have a stronger correlation than ^{87}Br, which is surprising considering its long half-life causes group one DNP group parameters to be more sensitive to it. However, differences in the emission probabilities, half-lives, and cumulative fission yields can cause DNPs such as ^{137}I to contribute more to DNP group two. For instance, Section III.A shows that the PCC for the group two half-life τ_2 from

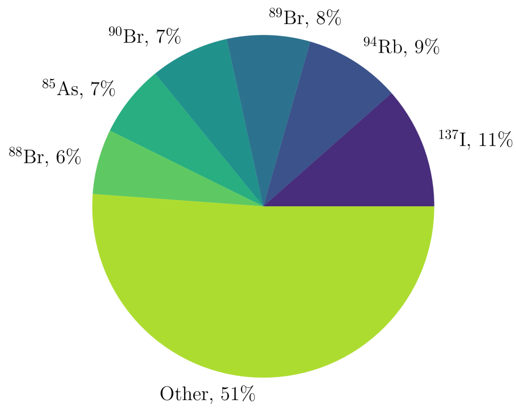


Fig. 6. The fraction of the delayed neutron yield from particular DNPs from ENDF/B-VII.1 nuclear data. The six DNPs with the largest delayed neutron yield contribute approximately 50% of the total delayed neutrons per fission event. The remaining 212 DNPs provide approximately 50% of the total delayed neutron yield.

TABLE VI
ENDF/B-VII.1 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0191 ± 0.0011	$[-]$
$\nu_d(K)$	0.0191 ± 0.0009	$[-]$
$\Delta\nu_d$	4.36 ± 135.24	$[pcm]$
$\bar{\tau}(I)$	7.2 ± 0.3	$[s]$
$\bar{\tau}(K)$	7.2 ± 0.3	$[s]$
$\Delta\bar{\tau}$	0.01 ± 0.4	$[s]$

TABLE VII
ENDF/B-VII.1 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{137}I	4.643721
^{137}Sb	4.612421
^{100}Rb	4.323455
^{87}Br	3.604110
^{85}As	3.025537
^{88}Br	2.177990
^{131}Cd	1.889176
^{88}As	1.880888
^{102}Y	1.311140
^{89}Br	1.242374

TABLE VIII
ENDF/B-VII.1 nuclear data ten largest uncertainty-weighted PCCs. Bold DNPs are present in Table VII as well, indicating they have a strong effect on DNP group parameters.

Nuclide	U_i
^{100}**Rb**	**1.365283**
^{137}**Sb**	**1.181643**
^{100}Kr	0.429832
^{99}Kr	0.370420
^{102}**Y**	**0.320689**
^{87}As	0.307750
^{124}Ag	0.275877
^{124}Pd	0.254152
^{84}As	0.249879
^{147}Xe	0.241646

changes in the ^{137}I half-life is 0.726; but that same value calculated using the ENDF/B-VII.1 nuclear data is 0.922.

Table VIII shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k. There are several DNPs that exist in both Table VII and VIII. These DNPs have a strong effect on the DNP group parameters and have large relative uncertainties. This means that, for ENDF/B-VII.1, these DNPs will lead to greater uncertainty in the calculated DNP group parameters and calculated delayed neutron emission rates.

Figure 7 gives a more detailed view on which DNP values specifically lead to the large U_i values. This offers slightly more insight than Table VIII, which sums over the various DNP values. These results indicate that the half-lives are the primary sources of uncertainty that most affect the DNP group parameters. The top two DNP in Table VIII appear twice in this figure, as they

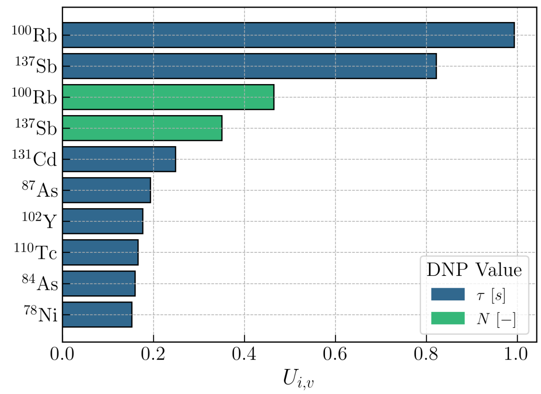
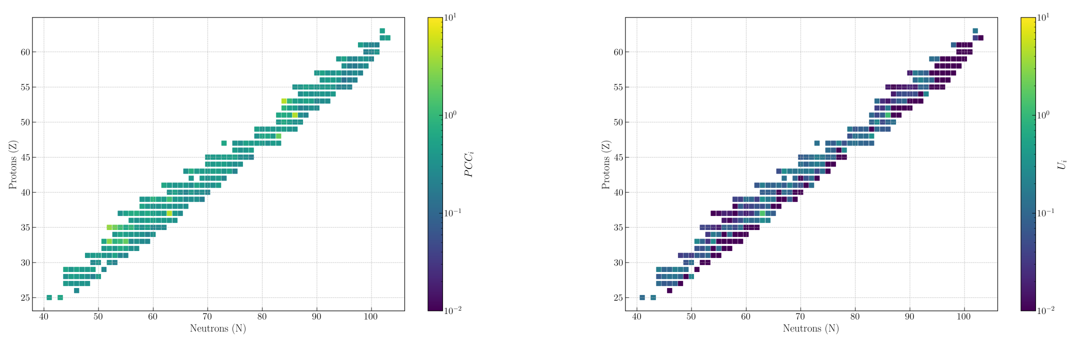


Fig. 7. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from ENDF/B-VII.1 nuclear data.

have contributions from both the half-life nuclear data and their concentration. The nuclear data for ^{100}Rb and ^{137}Sb are the most problematic in ENDF/B-VII.1 when calculating DNP group parameters.

Figure 8 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i, because all of the relative uncertainties are less than 1. There are two points of interest that have large PCC_i and U_i values, ^{100}Rb and ^{137}Sb, which can be seen on both plots.



(a) The PCC_i, or total PCC, calculated for each DNP.

(b) The U_i, or total uncertainty-weighted PCC, for each DNP.

Fig. 8. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using ENDF/B-VII.1 nuclear data.

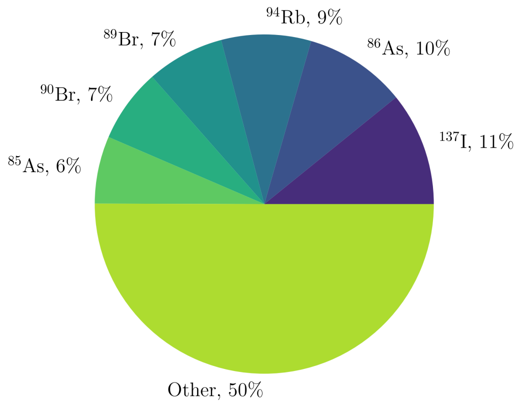


Fig. 9. The fraction of the delayed neutron yield from particular DNPs from ENDF/B-VIII.0 nuclear data. The six DNPs with the largest delayed neutron yield contributions approximately 50% of the total delayed neutrons per fission event. The remaining 271 DNPs provide approximately 50% of the total delayed neutron yield.

III.C. ENDF/B-VIII.0

Figure 9 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). There are 277 DNPs in ENDF/B-VIII.0, 59 more than in ENDF/B-VII.1. Although Figures 9 and 6 are similar, the main difference is the appearance of ^{86}As in the top six DNP yields.

Table IX shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general behavior of the delayed neutrons. However, the average half-life differs by approximately 0.3 seconds from Section III.B, while the total delayed neutron yield differs by approximately 100 pcm. These differences put both results within uncertainty bounds of each other, though the ENDF/B-VIII.0 total delayed neutron yield and average half-life have even larger differences from the values calculated using the IAEA beta-delayed neutron emission database and JENDL-5 than the ENDF/B-VII.1 data.

Table X shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k. These results are similar to those presented in Section III.B using ENDF/B-VII.1 data. The main changes are due to the presence of ^{86}As and ^{87}As, which were not present in the ten largest PCCs for ENDF/B-VII.1 data.

TABLE IX
ENDF/B-VIII.0 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0202 ± 0.0011	$[-]$
$\nu_d(K)$	0.0203 ± 0.0009	$[-]$
$\Delta\nu_d$	4.64 ± 142.02	$[pcm]$
$\bar{\tau}(I)$	6.9 ± 0.3	$[s]$
$\bar{\tau}(K)$	6.9 ± 0.3	$[s]$
$\Delta\bar{\tau}$	0.01 ± 0.4	$[s]$

TABLE X
ENDF/B-VIII.0 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{137}Sb	5.052885
^{100}Rb	4.837795
^{137}I	4.468315
^{87}Br	3.350813
^{85}As	2.685888
^{88}As	2.107151
^{88}Br	2.093959
^{131}Cd	1.988529
^{86}As	1.862035
^{87}As	1.472684

Table XI shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k. The results are very similar to those shown in Table VIII, though ^{87}As has taken the place of ^{102}Y for having a large PCC and relative uncertainty, appearing in both Table X and XI. Another new addition is ^{131}Cd, which does not have any differences in cumulative fission yield, emission probability, or half-life between ENDF/B-VII.1 and ENDF/B-VIII.0. Instead the PCCs increased because of differences in other DNPs, as can be seen by comparing the ^{131}Cd ENDF/B-VIII.0 PCC_i of 1.989 to the ENDF/B-VII.1 value of 1.889.

Figure 10 gives a more detailed view on which DNP values specifically lead to the large U_i values. This offers slightly more insight than Table XI, which sums over the various DNP values. The top four $U_{i,v}$ values are identical as those in Figure 7, indicating nuclear data for ^{100}Rb and ^{137}Sb remain problematic in ENDF/B-VIII.0 when calculating DNP group parameters.

Figure 11 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i, because all of the relative uncertainties

TABLE XI
ENDF/B-VIII.0 nuclear data ten largest uncertainty-weighted PCCs. Bold DNPs are present in Table X as well, indicating they have a strong effect on DNP group parameters.

Nuclide	U_i
^{100}Rb	**1.510501**
^{137}Sb	**1.274605**
^{87}As	**0.380737**
^{99}Kr	0.355989
^{102}Y	0.291115
^{84}As	0.268318
^{131}Cd	**0.253599**
^{98}Y	0.245287
^{104}Nb	0.240678
^{124}Ag	0.240391

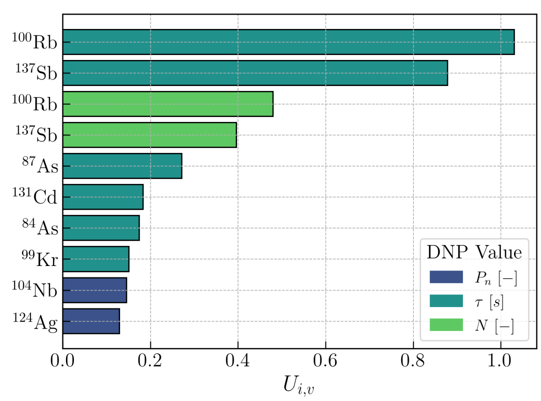
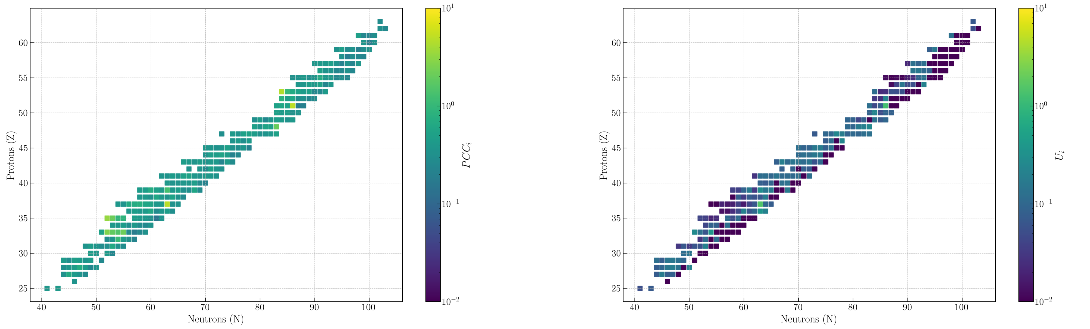


Fig. 10. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from ENDF/B-VIII.0 nuclear data.

are less than 1. The same points of interest as in Figure 8 are still relevant: ^{100}Rb and ^{137}Sb.



(a) The PCC_i, or total PCC, calculated for each DNP.

(b) The U_i, or total uncertainty-weighted PCC, for each DNP.

Fig. 11. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using ENDF/B-VIII.0 nuclear data.

III.D. JEFF-3.1.1

Figure 12 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). There are only 136 DNPs present in JEFF-3.1.1, and they have a total delayed neutron yield of 0.01477. This is much smaller than the total delayed neutron yields from other datasets, and thus leads each DNP to have a larger relative contribution to the delayed neutron yield.

Table XII shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general behavior of the delayed neutrons. Although the total delayed neutron yield is smaller than other datasets, the average half-life is longer, indicating that the JEFF-3.1.1 data may be missing or under-predicting concentrations of short-lived DNP in particular.

Table XIII shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k. The top two DNPs, ^{137}I and ^{87}Br, strongly affect groups two and one as previously discussed, which is why they have the largest PCCs. Section III.A has a more detailed discussion on what effects increase PCCs.

Table XIV shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k. There are several

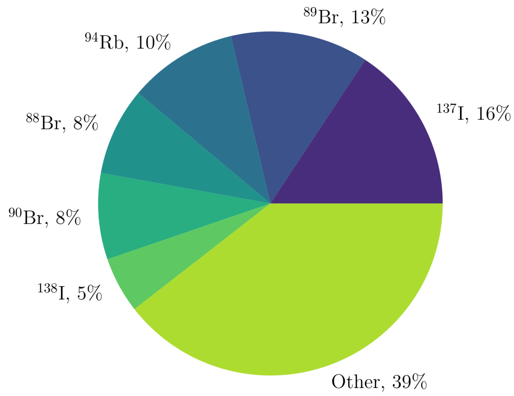


Fig. 12. The fraction of the delayed neutron yield from particular DNPs from JEFF-3.1.1 nuclear data. The five DNPs with the largest delayed neutron yield contributions approximately 56% of the total delayed neutrons per fission event. The remaining 131 DNPs provide approximately 44% of the total delayed neutron yield.

TABLE XII
JEFF-3.1.1 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0148 ± 0.0008	$[-]$
$\nu_d(K)$	0.0148 ± 0.0007	$[-]$
$\Delta\nu_d$	1.25 ± 108.88	$[pcm]$
$\bar{\tau}(I)$	9.2 ± 0.4	$[s]$
$\bar{\tau}(K)$	9.2 ± 0.4	$[s]$
$\Delta\bar{\tau}$	0.0 ± 0.6	$[s]$

TABLE XIII
JEFF-3.1.1 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{137}I	3.619275
^{87}Br	3.498093
^{88}Br	3.323438
^{138}I	2.993978
^{96}Rb	2.727061
^{89}Br	2.304603
^{98}Y	2.055720
^{94}Rb	1.959133
^{91}Br	1.919183
^{90}Br	1.886707

TABLE XIV
JEFF-3.1.1 nuclear data ten largest uncertainty-weighted PCCs. Bold DNPs are present in Table XIII as well, indicating they have a strong effect on DNP group parameters.

Nuclide	U_i
^{98}Y	**0.387086**
^{93}Br	0.237471
^{147}Xe	0.214894
^{96}Rb	**0.176298**
^{138}Te	0.171012
^{94}Br	0.152192
^{91}Br	**0.148399**
^{158}Nd	0.144933
^{137}I	**0.141580**
^{111}Mo	0.134707

DNPs that appear in Table XIV that also appear in Table XIII, indicated by the bold values. This highlighting indicates DNPs that strongly affect the DNP group parameters, meaning that, although their relative uncertainties may not be as large as the other DNPs in the table, the uncertainties they do have lead to large differences in the DNP group parameters and calculated delayed neutron emission rates.

Figure 13 gives a more detailed view on which DNP values specifically lead to the large U_i values. This offers slightly more insight than Table XIV, which sums over the various DNP values. For instance, these results show that ^{98}Y has a large contribution of its uncertainty-weighted sensitivity come from emission probability uncertainty, with a smaller contribution from its half-life uncertainty. It can also be seen that ^{137}I is not in Figure 13 even though it is in Table XIV, indicating that its uncertainty-weighted sensitivity comes from the sum of several smaller components rather than one large term. Specifically, 54.5% of its U_i comes from emission probability uncertainty, 42.8% comes from concentration uncertainty, and the remainder comes from half-life uncertainty.

Figure 14 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i, because all of the relative uncertainties are less than 1. There are similar patterns as discussed in Section III.A, with horizontal peaks in the PCC_i for bromine, rubidium, and iodine.

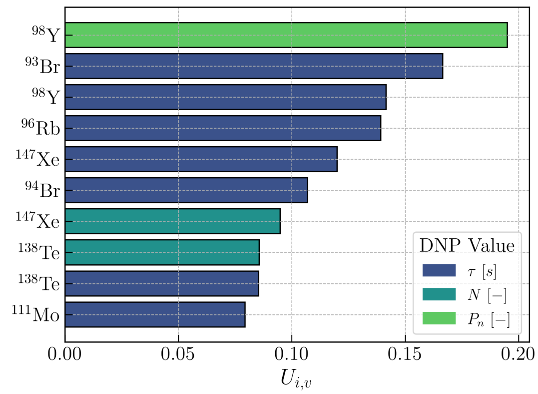
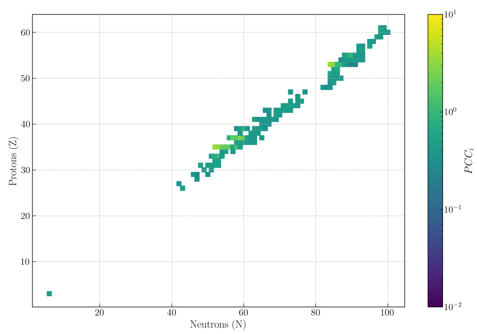
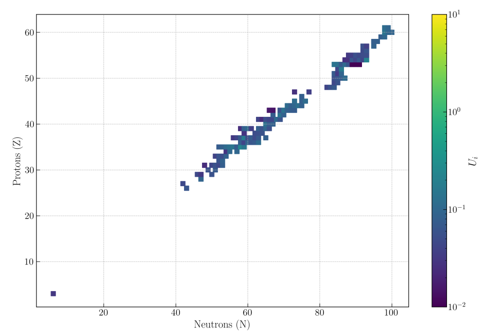


Fig. 13. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from ENDF/B-VII.1 nuclear data.



(a) The PCC_i, or total PCC, calculated for each DNP.



(b) The U_i, or total uncertainty-weighted PCC, for each DNP.

Fig. 14. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using JEFF-3.1.1 nuclear data.

III.E. JENDL-5

Figure 15 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). There are 257 DNPs present in JENDL-5, and they have a total delayed neutron yield of 0.0160. This is consistent with the results in Section III.A, as the only difference between these datasets is the emission probability and half-life data.

Table XV shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The results agree well with the data in Section III.A.

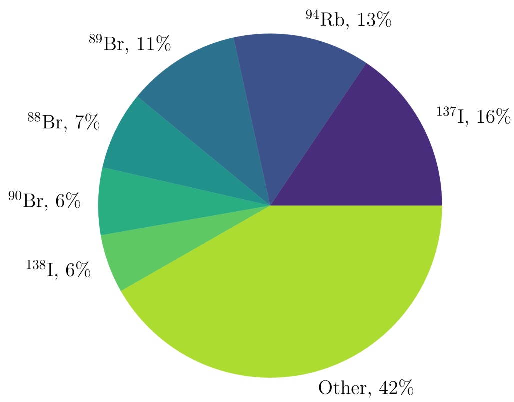


Fig. 15. The fraction of the delayed neutron yield from particular DNPs from JENDL-5 nuclear data. The five DNPs with the largest delayed neutron yield contributions approximately 52% of the total delayed neutrons per fission event.

TABLE XV
JENDL-5 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0160 ± 0.0009	$[-]$
$\nu_d(K)$	0.0160 ± 0.0008	$[-]$
$\Delta\nu_d$	0.09 ± 122.31	$[pcm]$
$\bar{\tau}(I)$	9.1 ± 0.5	$[s]$
$\bar{\tau}(K)$	9.1 ± 0.5	$[s]$
$\Delta\bar{\tau}$	0.0 ± 0.6	$[s]$

TABLE XVI
JENDL-5 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{87}Br	3.788538
^{88}Br	3.405082
^{137}I	3.373592
^{96}Rb	3.230474
^{97}Rb	2.406196
^{138}I	2.209843
^{94}Rb	2.202741
^{89}Br	2.123136
^{91}Br	1.779122
^{90}Br	1.733836

TABLE XVII
JENDL-5 nuclear data ten largest uncertainty-weighted PCCs.

Nuclide	U_i
^{102}Sr	-
^{142}Sb	0.283440
^{99}Kr	0.279344
^{105}Zr	0.270256
^{116}Rh	0.269286
^{103}Zr	0.267184
^{124}Ag	0.228787
^{68}Co	0.224881
^{134}Sn	0.207609
^{66}Mn	0.199226

Table XVI shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k. The same DNPs are in Table IV, though the exact values and order is slightly different due to half-life and emission probability differences.

Table XVII shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k. Four out of ten of these DNPs are also in Table V, though there are differences due to the emission probability and half-life differences. One notable difference is ^{102}Sr, which has a U_i value of 3.63×10^{13}. This value is significantly larger than any other value from any other dataset because it has an emission probability of $(6.245 \times 10^{-17}) \pm 0.02$. This means that the relative uncertainty in Equation (17) works out to be 3.2×10^{14}, which is exceedingly large. To continue conducting the analysis, the contribution of ^{102}Sr is ignored, treating the emission probability as 0 for this analysis.

Figure 16 gives a more detailed view on which DNP values specifically lead to the large

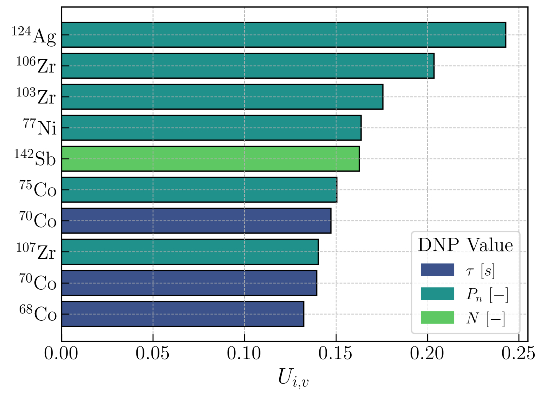
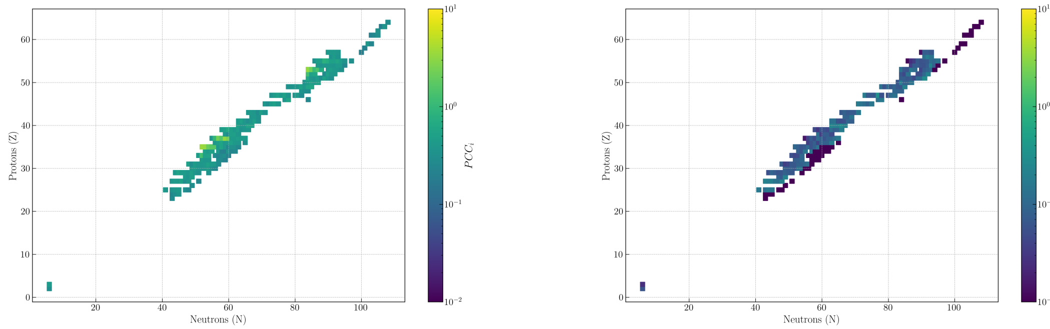


Fig. 16. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from ENDF/B-VII.1 nuclear data.

U_i values. This offers slightly more insight than Table XVII, which sums over the various DNP values. These uncertainty-weighted sensitivities are closer to the ideal configuration: values are close to each other, and the uncertainties in half-lives, emission probability, and concentration each contribute some amount. There are no clear outliers in the data, though, similar to the results in Section III.A, the largest uncertainty-weighted sensitivity contributions come from emission probabilities, indicating a potential area of improvement.

Figure 17 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i, because all of the relative uncertainties are less than 1. There are similar patterns as discussed in Section III.A, with horizontal peaks in the PCC_i for bromine, rubidium, and iodine.



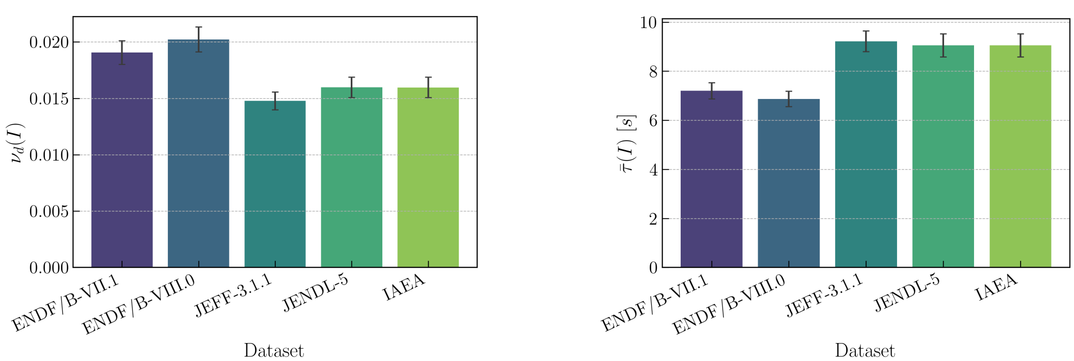
(a) The PCC_i, or total PCC, calculated for each DNP.

(b) The U_i, or total uncertainty-weighted PCC, for each DNP.

Fig. 17. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using JENDL-5 nuclear data.

III.F. Hybrid

The dataset used has a large affect on the total delayed neutron yield and average half-life values. Figure 18 shows how the previously discussed datasets compare to each other. These results show that the ENDF/B-VII.1 and ENDF/B-VIII.0 datasets have delayed neutron yields $\nu_d(I)$ and average half-lives $\bar{\tau}(I)$ within uncertainty bounds of each other. However, the ENDF results are not consistent with the results from JEFF-3.1.1, JENDL-5, and the IAEA dataset using JENDL-5 cumulative fission yields.

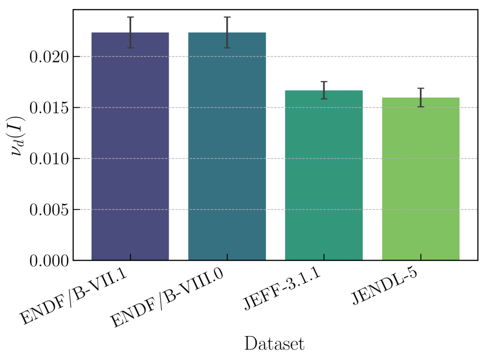


(a) The total delayed neutron yield.

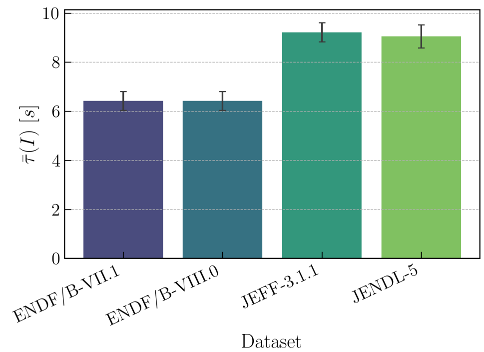
(b) The average half-life.

Fig. 18. Bar charts comparing the total delayed neutron yield $\nu_d(I)$ and average half-lives $\bar{\tau}(I)$ from the previously discussed datasets.

Figure 19 shows how the cumulative fission yields from the different datasets affect the results when the IAEA beta-delayed neutron emission database half-lives and emission probabilities are used. The results reflect those shown in Figure 18, where the JENDL-5 and JEFF-3.1.1 results are within uncertainty bounds while the ENDF results are within uncertainty bounds, but the two sets are not within uncertainty bounds of each other.



(a) The total delayed neutron yield.

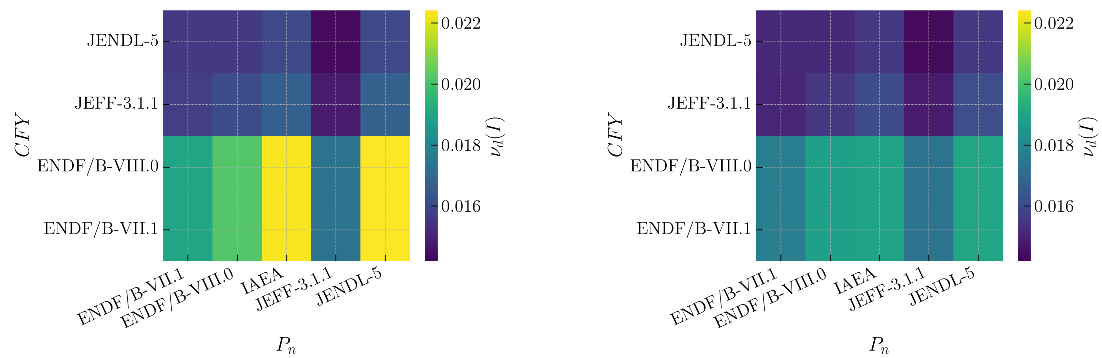


(b) The average half-life.

Fig. 19. Bar charts comparing the total delayed neutron yield $\nu_d(I)$ and average half-lives $\bar{\tau}(I)$ using the IAEA beta-delayed neutron emission database half-lives and emission probabilities for differing cumulative fission yield datasets.

However, it is also possible to combine datasets, mixing the cumulative fission yields, emission probabilities, and half-lives. Figure 20 shows heatmaps of the total delayed neutron yield $\nu_d(I)$ plotted against different cumulative fission yield and emission probability datasets. Although it is expected that the half-life dataset should not have an effect, only the nuclides present in all three datasets is included. This means that using the JEFF-3.1.1 half-lives results in a decrease to the total delayed neutron yield due to missing DNPs present in the ENDF datasets. Interestingly, this has a larger effect on the ENDF cumulative fission yield calculated delayed neutron yields compared to those from JEFF-3.1.1 and JENDL-5. This is because the ENDF data has large PCCs for certain DNPs, such as ^{88}As, that are not included in JEFF-3.1.1. This indicates the cumulative fission yields for these DNPs may be over-reported in ENDF data compared to JENDL-5 and JEFF-3.1.1.

It appears in Figure 20 that the cumulative fission yield differences are what drive the differences in the total delayed neutron yields, with smaller differences from the emission probabilities. This is useful to note, as based on the uncertainty-weighted PCCs calculated in Sections III.B and III.C, the emission probability data is the data in most need of updating. This is because the ENDF data has smaller relative uncertainties for the cumulative fission yields even though they do not agree well with the JEFF-3.1.1 and JENDL-5 cumulative fission yields.



(a) The total delayed neutron yield using ENDF/B-VII.1 half-lives.

(b) The total delayed neutron yield using JEFF-3.1.1 half-lives.

Fig. 20. Heatmaps of the total delayed neutron yield $\nu_d(I)$ calculated using various datasets for the cumulative fission yield, emission probability, and half-life.

Table XVIII shows a small collection of total delayed neutron yields from the literature calculated using summation approaches or from experimental data compare with the results from this work. There are many existing analyses that calculate and measure the total delayed neutron yield, with various collections available in the literature [2, 3, 7, 26, 27, 19]. Generally, there has been increasing agreement over time that the delayed neutron yield for thermal fission of ^{235}U is approximately 0.01625 ± 0.01 [28]. Among the values shown in Table XVIII and based on the agreed upon ν_d value, the outliers are the ENDF/B-VII.1 and ENDF/B-VIII.0 ν_d values.

III.F.1. Nuclear Data Comparisons

The results thus far have indicated potential discrepancies in the ENDF cumulative fission yields for the DNPs. To investigate this, cumulative fission yield differences between JENDL-5 and ENDF/B-VIII.0[e] are compared for the DNPs with the largest delayed neutron yield $\nu_d(I)$ contribution differences. This analysis showed there are two DNPs alone that account for 67% of the delayed neutron yield difference: ^{86}As and ^{137}Sb. The ^{86}As yield difference primarily comes from the cumulative fission yield difference, which is 540 ± 90 *pcm* larger in ENDF/B-VIII.0 than JENDL-5, and leads to a delayed neutron yield difference of 191 ± 32 *pcm*. The discrepancy between the ^{86}As in ENDF data has been reported on previously for ENDF/B-VII.0 and ENDF/B-VII.1, respectively [8, 6]. This yield difference is smaller in ENDF/B-VII.1, but that is because the

[e] Analysis with ENDF/B-VIII.1 gives the same results.

TABLE XVIII
Total delayed neutron yields ν_d from 100 thermal fissions of ^{235}U.

Data	ν_d $[-]$
JEF-2.2 [29, 7]	1.708 ± 0.12
JEF-3.1.1 [22, 7]	1.477 ± 0.08
ENDF/B-VII.1 [12, 7]	1.905 ± 0.10
Keepin et al. [24]	1.580 ± 0.07
Charlton et al. [30]	1.577 ± 0.03
Saleh et al. [31]	1.645 ± 0.11
Synetos and Williams [25]	1.510 ± 0.06
Tuttle [32]	1.700 ± 0.05
Waldo et al.[19]	1.670 ± 0.07
Rudstam et al. [3]	1.620
IAEA/JENDL-5 [11, 15]	1.596 ± 0.09
ENDF/B-VII.1 [12]	1.906 ± 0.11
ENDF/B-VIII.0 [14]	2.023 ± 0.11
ENDF/B-VIII.1 [33]*	2.023 ± 0.11
JEFF-3.1.1 [22]	1.477 ± 0.08
JENDL-5 [15]	1.597 ± 0.09

* ENDF/B-VIII.1 did not resolve the delayed neutron yield discrepancy present in the previous ENDF versions.

emission probability in ENDF/B-VII.1 is 0.12 instead of 0.36 as it is in ENDF/B-VIII.0.

The ^{137}Sb yield difference has several contributing components: the emission probability in JENDL-5 is 0.49 ± 0.08, while the emission probability in ENDF/B-VIII.0 is 1.0 ± 0.0[f]; the half-life in JENDL-5 is 0.94 ± 0.01 seconds, while the half-life in ENDF/B-VIII.0 is 0.45 ± 0.05 seconds; and finally the cumulative fission yield in JENDL-5 is $(2.4 \pm 0.9) \times 10^{-5}$, while the cumulative fission yield in ENDF/B-VIII.0 is $(7 \pm 5) \times 10^{-4}$. There are other DNPs that contribute to the total delayed neutron yield discrepancy between ENDF and the other data sets, such as the cumulative fission yield of ^{85}As, the cumulative fission yield of ^{90}Br, all of the data for ^{86}Ge, the cumulative fission yield of ^{94}Rb, the cumulative fission yield of ^{137}I, the cumulative fission yield and half-life of ^{98m}Y, and others. Of the DNPs listed, only ^{94}Rb and ^{137}I have larger yields in JENDL-5 than in ENDF/B-VIII.0; most of the differences in the nuclear data for DNPs lead to ENDF datasets having a larger delayed neutron yield.

For summation calculations, microscopic evaluations of the DNP group parameters, or any simulations of delayed neutron emission rates using individual DNP, these differences in the nuclear data are relevant to consider and account for when using ENDF data. The difference between

[f] No uncertainty is given, so an uncertainty of 1×10^{-12} is used.

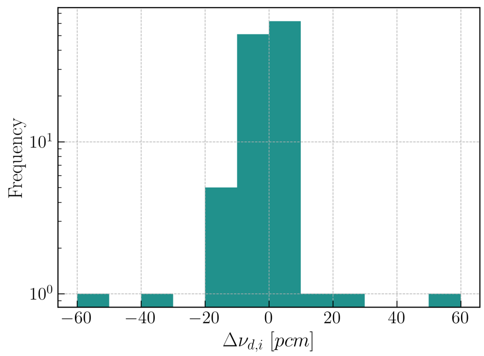
JENDL-5 and JEFF-3.1.1 was also investigated to determine which DNPs have the largest effect leading to JEFF-3.1.1 under-predicting the total delayed neutron yield. There is a 120 *pcm* difference between JENDL-5 and JEFF-3.1.1 total delayed neutron yields. As previously discussed in Section III.D, JEFF-3.1.1 has only 136 DNPs in its dataset, and 123 of those are present in JENDL-5. Those 123 that are in common have a total delayed neutron yield contribution in JEFF-3.1.1 that is 76 *pcm* less than the same DNPs in JENDL-5. The remaining 44 *pcm* delayed neutron yield difference between JENDL-5 and JEFF-3.1.1 comes from the DNPs the datasets do not have in common, as JENDL-5 has 257 DNP with half-life, emission probability, and cumulative fission yield data while JEFF-3.1.1 has only 136.

The difference between the JEFF-3.1.1 and JENDL-5 total delayed neutron yields does not come primarily come from one or two DNPs, as was the case for the ENDF data difference. The top four DNPs with the largest yield differences are ^{94}Rb, ^{98m}Y, ^{85}As, and ^{89}Br which are all present in both datasets and have both negative and positive differences[g]. Instead, the difference comes from many small ν_d contributions from JEFF-3.1.1 that are smaller than those in JENDL-5.

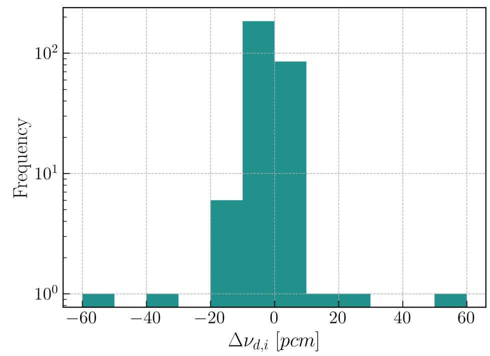
These small differences are displayed in Figure 21, which shows the JENDL-5 delayed neutron yields subtracted from the JEFF-3.1.1 delayed neutron yields using histograms with bin widths of 10 *pcm*. Figure 21(a) shows the histogram for the 123 DNPs present in both datasets. This histogram shows in the two largest bins of 0 to 10 *pcm* and -10 to 0 *pcm* there are 62 DNPs with an average $\Delta\nu_{d,i}$ of 0.90 *pcm* contributing a total of 55.5 *pcm*, while there are 51 DNPs with an average $\Delta\nu_{d,i}$ of -1.22 *pcm* contributing a total of -62.2 *pcm*, respectively. The average $\Delta\nu_{d,i}$ when using the 123 DNPs present in both datasets is -0.62 *pcm*, which comes from many small contributions as well as the seven DNPs with a $\Delta\nu_{d,i}$ less than -10 *pcm*, though this is slightly offset by the three DNPs with a $\Delta\nu_{d,i}$ greater than 10 *pcm*. Figure 21(b) shows the histogram for the 280 unique nuclides between both datasets. This histogram shows in the two largest bins of 0 to 10 *pcm* and -10 to 0 *pcm* there are 85 DNPs with an average $\Delta\nu_{d,i}$ of 0.66 *pcm* contributing a total of 55.9 *pcm*, while there are 184 DNPs with an average $\Delta\nu_{d,i}$ of -0.49 *pcm* contributing a total of -90.7 *pcm*, respectively. The average $\Delta\nu_{d,i}$ when using the 280 DNPs present in either dataset is -0.43 *pcm*, which comes from many small contributions as well as the seven DNPs with a $\Delta\nu_{d,i}$ less than -10 *pcm*, though this is slightly offset by the three DNPs with a $\Delta\nu_{d,i}$ greater

[g] ^{94}Rb is larger in JENDL-5 by 54 *pcm*, ^{98m}Y is larger in JEFF-3.1.1 by 52 *pcm*, ^{85}As is larger in JENDL-5 by 39 *pcm*, and ^{89}Br is larger in JEFF-3.1.1 by 21 *pcm*.

than 10 *pcm*. The average is smaller in magnitude compared to using only the common DNPs because there are many additional smaller contributions present when account for the many less impactful DNPs present in JENDL-5 that are not present in JEFF-3.1.1.



(a) The JEFF-3.1.1 DNP ν_{d_i} minus the JENDL-5 DNP $\nu_{d,i}$ for DNPs present in both datasets.



(b) The JEFF-3.1.1 DNP ν_{d_i} minus the JENDL-5 DNP $\nu_{d,i}$ for every DNP present in either dataset.

Fig. 21. These histograms show the difference in the individual DNP yield contribution $\Delta\nu_{d,i}$ between the JEFF-3.1.1 dataset and the JENDL-5 dataset using bin widths of 10 *pcm*. Using all DNP present in either dataset results in an average delayed neutron yield contribution difference $\Delta\nu_{d,i}$ of -0.43 *pcm*, while using only DNP present in both datasets results in an average delayed neutron yield contribution difference $\Delta\nu_{d,i}$ of -0.62 *pcm*.

IV. CONCLUSIONS

This work presented analyses using the microscopic approach to compare DNP nuclear data between the IAEA beta-delayed neutron emission database, ENDF/B-VII.1, ENDF/B-VIII.0, JEFF-3.1.1, JENDL-5, and hybrid datasets. These results, when compared with other summation calculations in the literature and experimental data, indicate that each dataset has specific DNPs that have large relative uncertainties with a greater impact on the calculation of DNP group parameters and delayed neutron emission rates than other DNPs.

Large differences in ENDF datasets that lead to discrepancies when comparing results with other summation calculations and experimental data are presented, specifically indicating problematic nuclides of ^{85}As, ^{86}As, ^{100}Rb, ^{137}Sb. The under-prediction of JEFF-3.1.1 for calculation of the total delayed neutron yield is also presented, indicating that the largest individual differences come from ^{94}Rb, ^{98m}Y, ^{85}As, and ^{89}Br; however, the majority of the difference comes from small contributions from many DNPs both present in the dataset and lack of contribution from

DNPs missing from the dataset. The JENDL-5 dataset is shown to have good agreement with experiments and other calculations in the literature, including when using the IAEA beta-delayed neutron emission database. The DNPs with the largest uncertainty-weighted sensitivities that affect delayed neutron emission rates and calculated DNP group parameters are provided for each dataset evaluated in this work. A discussion on hybrid datasets is also provided, which provides evidence that cumulative fission yields in ENDF datasets contribute to larger delayed neutron yields than other results in the literature.

The results in this work offer a new perspective for identifying DNPs in nuclear datasets that are in need of reduced uncertainties or lead to discrepancies with other results in the literature. These results can be leveraged to resolve nuclear data discrepancies and conduct experiments to improve the most relevant DNP nuclear data. Through these targeted improvements to the nuclear data, future calculations using the individual DNP nuclear data can be improved more rapidly. These improvements will improve the accuracy of transient simulations for nuclear reactors and enable more in-depth analyses on DNP behavior.

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